

Siddharth (Sid) Pandit

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Education

Oklahoma School of Science and Mathematics

High School Diploma – Rising Senior

Expected May 2027

Oklahoma City, OK

Experience

Research Assistant | *Mobility Intelligence Lab (MiLa), University of Oklahoma*

May 2026 - Present

Embedded Systems, Autonomous Driving, Computer Vision

Norman, Oklahoma

- Conduct autonomous driving research under Dr. Bin Xu, dedicating 50+ hours/week to integrating embedded systems, CAN communication, and camera-based perception.
- Designed, assembled, and validated custom PCBs integrating 25+ electronic components, and developed object detection pipelines in Python that processed 50+ GB of real-world driving data.

FIRST Robotics | *Francis Tuttle Technology Center*

Dec 2024 - Apr 2025

Autonomous Systems, Path Planning, Sensor Integration, WPILib

- Developed and optimized Java/WPILib code for autonomous routines, implementing path planning logic, closed-loop motion control, and real-time sensor integration for competition robots.

Smart Walking Cane | *Texas Instruments (TI)*

Oct 2024 - May 2025

Ultrasonic Sensor, Haptic Feedback, Engineering Design

Edmond, Oklahoma

- Designed and built an ultrasonic sensor-based prototype that processed noisy distance data within a 0.5 to 3-meter range in real time to generate reliable haptic feedback.
- Programmed distance-based vibration feedback that dynamically increased intensity as obstacles became closer, contributing to an Honorable Mention out of 400+ competing teams in the TI Codes Contest.

Engineering Education Research | *University of Oklahoma*

July 2024 - Dec 2024

Groq API Integration, MERN Stack Developer

Norman, Oklahoma

- Led development of an AI chatbot that helped engineering students explore career and academic paths, building an interactive web interface with Groq API integration and improving responses through iterative testing and feedback.

Skills

Programming Languages: Java, Python, JavaScript/TypeScript, C and C++

Embedded Systems & Hardware: Sensor integration, real-time signal processing, motor/actuator control, ultrasonic ranging, vibrational feedback systems; path planning logic, closed-loop control (WPILib), ESP32, CAN Bus, PCB Assembly & Soldering, Relay Control

Computer Vision & AI: Object Detection, OpenCV, PyTorch

Tools & Platforms: Git/GitHub, VS Code, Streamlit, local/browser-based data processing

Organizations

Technology Student Association (TSA) | Team Captain

Oct 2024 - June 2025

- Coordinated across Web Development, Data Science Analytics, Robotics, and Senior Solar Sprint competition projects development, testing, and competition preparation, contributing to state-level success in Webmasters and Robotics.

FIRST Robotics | Head Engineering Notebook Coordinator

Dec 2024 - Apr 2025

- Assigned and tracked team tasks to keep production on schedule, while compiling a detailed engineering portfolio that earned the Think Award for the region.

Volunteer Experience

Middle School Math Workshop | Mentor, 20+ hrs.

May - June 2025

- Mentored middle schoolers in a multi-day math workshop, providing one-on-one guidance and targeted problem-solving support that boosted engagement, comprehension, and confidence in mastering challenging concepts.

Honors and Awards

- BPA – 1st Place, Website Design (State & National)
- National Honor Society Member, 2024–2025

- TSA – 1st Place, Webmasters (Oklahoma State)
- Francis Tuttle Student Spotlight, October 2024

Relevant Coursework

- Data Structures I
- Calculus I
- AP CS Principles
- AP CS A
- Mechanics
- Object Oriented Programming
- General Physics